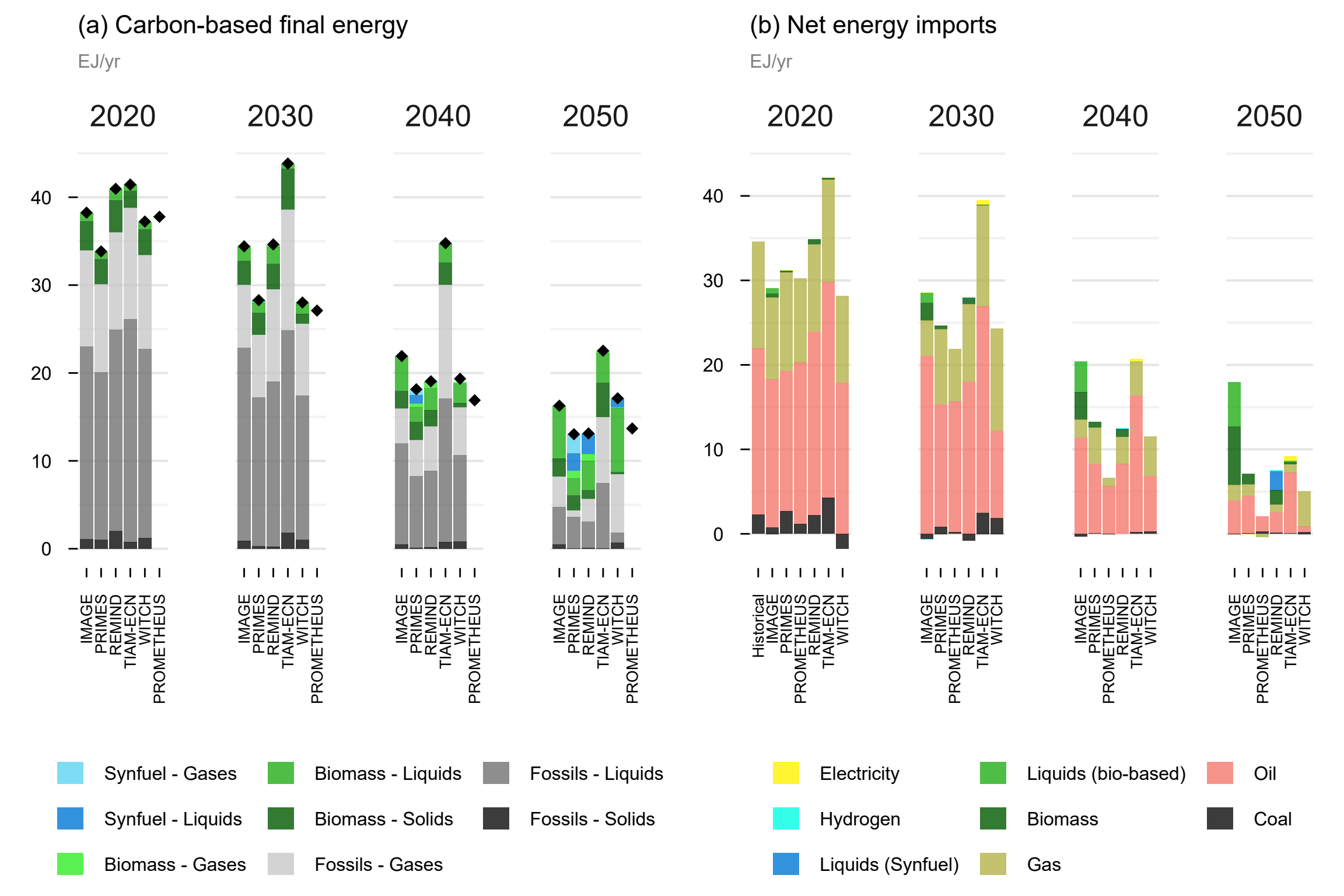
(a) Carbon-based final energy
EJ/yr
2020
2030
2040
2050
40
30
20
10
0
IMAGE
PRIMES
REMIND
TIAM-ECN
WITCH
PROMETHEUS
(b) Net energy imports
EJ/yr
Historical
Synfuel - Gases
Synfuel - Liquids
Biomass - Gases
Biomass - Liquids
Biomass - Solids
Fossils - Gases
Fossils - Liquids
Fossils - Solids
Electricity
Hydrogen
Liquids (Synfuel)
Liquids (bio-based)
Biomass
Gas
Oil
Coal